Proof The proof is very simple, and follows almost directly from our ergodicity assumption. If we assume that

$$\lim_{n\to\infty} (\mathcal{ST})^n \rho = \hat{\rho}_{ST},$$

for any initial distribution ρ, then we have

$$\hat{\rho}_{ST} = \lim_{n\to\infty} (\mathcal{ST})^n \hat{\rho}_{ST} = \lim_{n\to\infty} \mathcal{S} (\mathcal{TS})^{n-1} \mathcal{T} \hat{\rho}_{ST} = \mathcal{S}\hat{\rho}_{TS}.$$

A similar statement holds for $\hat{\rho}_{ST}$. □

If we return to our scheme ⟦OYZYO⟧, which has a simple inhomogeneity, then we can relate its invariant distribution to invariant distributions of schemes with a central OU update by writing its evolution operator as

$$e^{h\hat{\mathcal{L}}^\dagger_{[\![\mathrm{OYZYO}]\!]}} = \underbrace{e^{(h/2)\mathcal{L}^\dagger_{\mathrm{O}}} e^{(h/2)\mathcal{L}^\dagger_{Y}} e^{(h/2)\mathcal{L}^\dagger_{Z}}}_{\mathcal{S}} \underbrace{e^{(h/2)\mathcal{L}^\dagger_{Z}} e^{(h/2)\mathcal{L}^\dagger_{Y}} e^{(h/2)\mathcal{L}^\dagger_{\mathrm{O}}}}_{\mathcal{T}} = \mathcal{ST}.$$

By applying Lemma 7.1, we can equate the invariant distributions directly as

$$\hat{\rho}_{[\![\mathrm{ZYOYZ}]\!]} = \exp((h/2)\mathcal{L}^\dagger_Z) \exp((h/2)\mathcal{L}^\dagger_Y) \exp((h/2)\mathcal{L}^\dagger_{\mathrm{O}}) \hat{\rho}_{[\![\mathrm{OYZYO}]\!]}.$$

Rewriting our ansatz (7.27) in place of the distributions, and expanding the exponential series gives

$$\begin{aligned}
&\rho_\beta \left(1 + h^2 f_{2,[\![\mathrm{ZYOYZ}]\!]} + \ldots\right) \\
&\quad = \left(\mathrm{Id} + \frac{h}{2}\mathcal{L}^\dagger_{\mathrm{LD}} + \frac{h^2}{4}\mathcal{L}^\dagger_{1,[\![\mathrm{OYZ}]\!]} + \frac{h^2}{2}\mathcal{L}^\dagger_{\mathrm{LD}} + \ldots\right) \left(\rho_\beta \left(1 + h^2 f_{2,[\![\mathrm{OYZYO}]\!]} + \ldots\right)\right),
\end{aligned}$$

and hence equating powers we have

$$f_{2,[\![\mathrm{ZYOYZ}]\!]}\rho_\beta = f_{2,[\![\mathrm{OYZYO}]\!]}\rho_\beta + \frac{1}{4}\mathcal{L}^\dagger_{1,[\![\mathrm{OYZ}]\!]}\rho_\beta.$$

By virtue of (7.20), we have

$$\mathcal{L}^\dagger_{1,[\![\mathrm{OAB}]\!]}\rho_\beta = \frac{1}{2}\mathcal{L}^\dagger_H (g\rho_\beta) \qquad \mathcal{L}^\dagger_{1,[\![\mathrm{OBA}]\!]}\rho_\beta = -\frac{1}{2}\mathcal{L}^\dagger_H (g\rho_\beta),$$

where recall $g(\boldsymbol{q},\boldsymbol{p})\rho_\beta = \mathcal{L}^\dagger_{\mathrm{A}}\rho_\beta = \beta \boldsymbol{p}^T \boldsymbol{M}^{-1} \nabla U(\boldsymbol{q})\rho_\beta$. Computing

$$\mathcal{L}^\dagger_H (g\rho_\beta) = \beta \left(|\nabla U(\boldsymbol{q})|^2 - \boldsymbol{p}^T \nabla^2 U(\boldsymbol{q}) \boldsymbol{p}\right) \rho_\beta,$$

the correction for the schemes with a central O update can be expressed as

$$f_{2,[\![\text{ABOBA}]\!]} = f_{2,[\![\text{OBABO}]\!]} - \frac{\beta}{8}\left(|\nabla U(\boldsymbol{q})|^2 - \boldsymbol{p}^T\nabla^2 U\boldsymbol{p}\right), \tag{7.30a}$$

$$f_{2,[\![\text{BAOAB}]\!]} = f_{2,[\![\text{OABAO}]\!]} + \frac{\beta}{8}\left(|\nabla U(\boldsymbol{q})|^2 - \boldsymbol{p}^T\nabla^2 U\boldsymbol{p}\right). \tag{7.30b}$$

Thus we need only compute the solutions to the simpler PDE (7.29), in order to relate the solutions of the conjugate schemes with centralized OU update. However, although this simplifies our task greatly, finding the general solution to (7.29) is still very challenging, and obtaining a closed form expression for the correction function is nontrivial.

7.9.2 Expansion in Powers of γ^{-1}

Although we do not give the general analytical solution to the partial differential equation of (7.28) for each of the four symmetric Langevin schemes we are considering, we can easily find a solution in an important limiting case: the high friction regime. We consider expanding f_2 into a series involving the reciprocal friction $\varepsilon = 1/\gamma$, and assume that for large friction our ansatz remains sensible. We will denote

$$f_2(\boldsymbol{q},\boldsymbol{p}) = f_{2,0}(\boldsymbol{q},\boldsymbol{p}) + \varepsilon f_{2,1}(\boldsymbol{q},\boldsymbol{p}) + \varepsilon^2 f_{2,2}(\boldsymbol{q},\boldsymbol{p}) + \ldots \tag{7.31}$$

with each $\langle f_{2,i}\rangle = 0$. Dividing by the friction in (7.28) we obtain

$$\left(\bar{\mathcal{L}}_{\text{O}}^{\dagger} + \varepsilon\mathcal{L}_{H}^{\dagger}\right)(f_2\rho_\beta) = -\varepsilon\mathcal{L}_2^{\dagger}\rho_\beta,$$

where $\bar{\mathcal{L}}_{\text{O}}^{\dagger}$ is independent of the friction

$$\bar{\mathcal{L}}_{\text{O}}^{\dagger}\vartheta = \varepsilon\mathcal{L}_{\text{O}}^{\dagger}\vartheta = \nabla_p\cdot(\boldsymbol{p}\vartheta) + \frac{1}{\beta}\Delta_p\vartheta.$$

From our earlier computation in (7.29), we can write

$$\mathcal{L}_{2,[\![\text{OABAO}]\!]}^{\dagger}\rho_\beta = w(\boldsymbol{q},\boldsymbol{p})\rho_\beta,$$

for $w(\boldsymbol{q},\boldsymbol{p})$ independent of the friction

$$w(\boldsymbol{q},\boldsymbol{p}) = -\frac{\beta}{24}\left(6\boldsymbol{p}^T\left(\nabla^2 U\right)\nabla U - \nabla^3 U : \boldsymbol{p}\otimes\boldsymbol{p}\otimes\boldsymbol{p}\right).$$